

Definition

Subspace Topology

Let (X, T) be a topological space. Then (Y, T_Y) is a **subspace topology** if

- Y is a subset of X , and
- A set $U \subseteq Y$ is open if there is some $V \subseteq X$ open in X where $U = V \cap Y$. That is,

$$U \in T_Y \iff \exists V \in T \text{ st. } U = V \cap Y.$$

Note

We use the notation **open_X** to say that a set is open in a topological space X . If a set is open in a subspace topology, we may say the set is **relatively open**.

Proposition

A subspace topology satisfies the topological axioms.

Proof: Suppose (Y, T_Y) is a subspace topology of (X, T) . Then

- We have $\emptyset = \emptyset \cap Y$, and $\emptyset$ is open_X, so $\emptyset$ is open_Y.
- Also, $Y = X \cap Y$, and X is open_X, so Y is open_Y.
- If U_1 and U_2 are open_Y, then

$$U_1 = V_1 \cap Y \quad \text{and} \quad U_2 = V_2 \cap Y$$

for some V_1 and V_2 , both open_X. So $U_1 \cap U_2 = (V_1 \cap V_2) \cap Y$. We know $V_1 \cap V_2$ is open_X by the topological axioms, so $U_1 \cap U_2$ is open_Y.

- Suppose U_α are open_Y for α in some indexing set. Then $U_\alpha = V_\alpha \cap Y$ for some open_X set V_α , for every α . Hence

$$\bigcup U_\alpha = \bigcup (V_\alpha \cap Y) = \left(\bigcup V_\alpha \right) \cap Y.$$

Since $\bigcup V_\alpha$ is open_X, the union $\bigcup U_\alpha$ is open_Y. ■

Definition

Relatively Closed

Let $A \subseteq Y \subseteq X$ for a topological space (X, T) and subspace topology (Y, T_Y) . Then we say A is **closed_Y**, or **relatively closed**, if $Y \setminus A$ is open_Y.

Theorem

Suppose $A \subseteq Y \subseteq X$ for a topological space (X, T) and subspace topology (Y, T_Y) . Then A is closed_Y if and only if $A = B \cap Y$ for some B that is closed_X.

Proof: To begin, suppose A is closed_Y. Then $Y \setminus A$ is open_Y, so $Y \setminus A = V \cap Y$ for some V that is open_X. It follows that $A = Y \setminus V$. Furthermore, $A = (X \setminus V) \cap Y$, hence $X \setminus Y$ is closed_X. Therefore, A is a closed_X set intersected with Y . ■

Exercise

Finish the proof above by showing the converse.

Theorem

Suppose $A \subseteq Y \subseteq X$ and $b \in Y$ where (X, T) is a topological space with a subspace topology (Y, T_Y) . Then b is adherent_X to A if and only if b is adherent_Y to A .

Proof: Suppose b is adherent_X to A . Let U be an arbitrary open_Y neighborhood of b . So $U = V \cap Y$ for some V that is open_X . Hence V is an open neighborhood of a point adherent_X to A , meaning V meets A . Thus U meets A , as $U \cap A = V \cap A$. So b is adherent_Y to A .

Conversely, suppose b is adherent_Y to A . Let V be an open_X neighborhood of b . Then $V \cap Y$ is an open_Y neighborhood of b , so $V \cap Y$ meets A . Thus

$$V \cap A = V \cap Y \cap A \neq \emptyset.$$

So V meets A , meaning b is adherent_X to A . ■

Continuity in Topological Spaces

Definition

Continuous

Let $f : X \rightarrow Y$ be a function where (X, T) and (Y, U) are topological spaces. Then f is **continuous** if for every $V \subseteq Y$ that is open_Y , the preimage $f^{-1}(V)$ is open_X .

Definition

Continuous at a Point

Let $f : X \rightarrow Y$ where (X, T) and (Y, U) are topological spaces. Suppose $x \in X$ and its image $f(x) = y \in Y$.

Then f is **continuous at x** if for every open neighborhood V of y , $f^{-1}(V)$ is a neighborhood of x . That is, if V is an open neighborhood of y , then x is interior to $f^{-1}(V)$.

Note that the preimage of V need not be an open neighborhood.

Theorem

Suppose $f : X \rightarrow Y$ is a function over topological spaces (X, T) and (Y, U) . Then f is continuous if and only if it is continuous at every point.

Proof: Suppose f is continuous. Let $x \in X$ be arbitrary. Further, let V be an open neighborhood of $f(x)$. Then $f^{-1}(V)$ is open and contains x , so x is certainly interior to $f^{-1}(V)$.

Conversely, suppose f is continuous at every point. Let $V \subseteq Y$ be an arbitrary open set. We prove $f^{-1}(V)$ is open, one point at a time.

Let $x \in f^{-1}(V)$. Then $f(x) \in V$. So V is an open neighborhood of $f(x)$, and since f is continuous at x , we know x is interior to $f^{-1}(V)$. Thus $x \in U_x \subseteq f^{-1}(V)$ for some open set U_x . Then

$$f^{-1}(V) = \bigcup_{x \in f^{-1}(V)} U_x$$

is open by the topological axioms. ■

Theorem

If X , Y , and Z are topological spaces, and $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then $g \circ f : X \rightarrow Z$ is also continuous.

Proof: Let $V \subseteq Z$ be open. Then

$$(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V)).$$

We know $g^{-1}(V)$ is open since g is continuous. So $f^{-1}(g^{-1}(V))$ is open because f is also continuous. Therefore, $g \circ f$ is continuous. ■

Definition

Homeomorphism

Let X and Y be topological spaces. Then $f : X \rightarrow Y$ is a **homeomorphism** if

- f is injective
- f is surjective
- f is continuous
- $f^{-1} : Y \rightarrow X$ is continuous where f^{-1} is the inverse of f

Example

Without proof, here are some spaces that are homeomorphic. If X and Y are homeomorphic, we denote it with $X \cong Y$.

- $(0, 1) \cong \mathbb{R}$.
- $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cong \mathbb{R}$ with the $\tan$ function
- A donut and a coffee cup

Example

Suppose X and Y are subspaces of $\mathbb{R}$ where

$$X = [1, 2] \cup (3, 4] \quad \text{and} \quad Y = [1, 3].$$

Suppose further that $f : X \rightarrow Y$ is defined as

$$f(x) = \begin{cases} x & \text{if } x \in [1, 2] \\ x - 1 & \text{if } x \in (3, 4] \end{cases}$$

Then $f^{-1} : Y \rightarrow X$ would be defined as

$$f^{-1}(x) = \begin{cases} x & \text{if } x \in [1, 3] \\ x + 1 & \text{if } x \in (2, 3] \end{cases}$$

But f^{-1} is not continuous. So f is not a homeomorphism.